

CRISTOFF STAN

Senior Software Developer

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PROFESSIONAL SUMMARY

Innovative and detail-oriented Senior Software Developer with over 8 years of experience in developing immersive and scalable web applications using modern JavaScript frameworks. Specialized in 3D graphics programming with Three.js and React-Three-Fiber, with a strong background in performance optimization, rendering pipelines, and real-time interactivity. Experienced with GLTF/OBJ model handling, shader-based visual effects, and seamless integration of complex 3D experiences into production web environments. Passionate about delivering visually compelling user experiences while adhering to clean code principles and best development practices.

TECHNICAL SKILLS

- **3D & WebGL:** Three.js, React-Three-Fiber, Shader Material, WebGL2, post-processing, lighting systems
- **Frontend:** React, Next.js, TypeScript, JavaScript (ES6+), Redux, TailwindCSS, Vite, HTML5, CSS3
- **3D Assets:** GLTF/GLB, OBJ, DRACO, Equirectangular, 360° panorama integration
- **Rendering Optimization:** requestAnimationFrame, frustum culling, texture compression, lazy loading, memoization, Lod management
- **Backend:** Node.js, NestJS, Express.js, REST APIs
- **Database:** MySQL, PostgreSQL
- **Tooling:** Webpack, Babel, Git, Docker, Figma, Jira
- **Testing:** Cypress, Jest, React Testing Library

WORK EXPERIENCE

Sensidev

Senior Software Engineer

September 2021 – December 2024

Timișoara, Romania

- Developed highly interactive 3D web-based simulations using Three.js and React-Three-Fiber for real-estate and architectural visualizations with complex lighting and scene composition.
- Engineered custom GLSL shaders for dynamic material effects, including water, metal, and transparency, integrated within physically-based rendering (PBR) pipelines.
- Optimized rendering pipelines by implementing frustum culling, DRACO-compressed GLTF loading, and efficient texture atlases, reducing average scene load time by 45%.
- Collaborated with designers to incorporate equirectangular projection and 360° panoramas into immersive room tours accessible via desktop and mobile.
- Maintained reusable hooks and component systems that managed 3D model states, loading progress, and camera animations with React context and Zustand.

- Refactored legacy animation code to utilize requestAnimationFrame with dynamic frame interpolation for consistent frame rates across devices.
- Integrated and managed large-scale GLTF models using KTX2 texture compression and automated LOD generation for real-time web environments.
- Led peer reviews and mentored junior developers in 3D web technologies and frontend architecture, improving code quality and team scalability.

Syscom Digital

Software Engineer

October 2019 – August 2021

Bucharest, Romania

- Built marketing visualizers using Three.js and WebGL that allowed interactive product showcases with shader-based lighting, real-time color changes, and hotspots.
- Designed modular React components to handle user interaction logic within embedded 3D viewers with drag, zoom, and orbit controls.
- Enhanced application responsiveness through adaptive rendering strategies and scene graph pruning for mobile and low-performance devices.
- Managed multi-format 3D assets (OBJ, FBX, GLTF) and implemented a converter pipeline to compress and optimize asset delivery for web.
- Collaborated closely with backend engineers to design APIs for dynamic loading of textures, meshes, and metadata into scenes.
- Introduced automatic scene baking using Blender for light maps and AO to reduce GPU workload during runtime rendering.
- Deployed visual performance monitoring tools like stats.js and Spector.js to diagnose and resolve GPU bottlenecks.
- Worked in an Agile team delivering on tight marketing deadlines and iterating based on A/B test feedback and user tracking analytics.

Soft Build

Software Developer

June 2016 – July 2019

Timișoara, Romania

- Developed interactive floor planners and configurators for real-estate using Canvas2D and WebGL transitioning later to Three.js.
- Managed multi-layer rendering scenes for combining interactive 2D UI with embedded 3D previews of interior components.
- Built drag-and-drop modules for placing furniture models with snapping mechanics and real-time size adjustments.
- Implemented user-friendly UI using React and Material UI to control light intensity, camera switching, and view settings in scenes.
- Created unit tests for 3D component behavior and ensured visual consistency across browsers and screen sizes.
- Collaborated with graphics designers to integrate custom 3D exports from Cinema4D and Blender into optimized web models.
- Added analytics tracking to capture user interaction heatmaps and session replays for conversion optimization.
- Contributed to a modular design system to reuse building blocks like modal dialogs, panels, and scene managers.

Decima Digital

Software Developer

November 2015 – April 2016

Tallinn, Estonia

- Contributed to a web-based 3D retail product configurator leveraging Three.js to allow interactive exploration of modular products.
- Built a user customization panel using React with controls linked to real-time model updates in the viewport.
- Implemented orthographic camera switching and ambient occlusion shading for enhanced scene realism.
- Wrote scene loading and saving logic with JSON serialization of camera and object states.
- Assisted in implementing post-processing pipelines including SSAO and bloom using EffectComposer.
- Worked with WebSockets to enable real-time collaborative sessions between users and sales reps.
- Optimized texture loading strategies to improve load speed and reduce memory usage.
- Participated in sprint planning, QA testing, and UX refinement based on client feedback.

Borna Technology

Software Developer Intern

September 2014 – August 2015

Tallinn, Estonia

- Supported 3D team by assisting with asset pipeline development including scripting GLTF optimizations.
- Built internal tools to validate and preview models in a custom WebGL sandbox environment.
- Developed proof-of-concept panorama viewers using 360° equirectangular images and spherical mapping.
- Contributed to early-stage experiments with Three.js shaders to simulate different material properties.
- Assisted with debugging user interaction layers in VR viewer prototypes using cardboard/webVR APIs.
- Documented rendering workflows and asset requirements for engineering onboarding.
- Created test suites for verifying correct LOD level switching based on camera distance.
- Performed UI testing and implemented accessibility fixes in early versions of the 3D web player.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- ❖ Certified Three.js Developer (Three.js Journey by Bruno Simon)
- ❖ Advanced WebGL & Graphics Programming (Udemy)
- ❖ React Performance Optimization Masterclass (Frontend Masters)

PROJECTS

✓ **Virtual Apartment Tour Engine**

Developed a scalable virtual tour platform with React-Three-Fiber and DRACO-compressed GLTF models. Enabled seamless transition between rooms with animated camera movement, interactive object tooltips, and HDRI-based lighting.

✓ **360° Panorama Explorer**

Implemented a lightweight, mobile-friendly viewer for real-estate walkthroughs using equirectangular projection and gyro controls. Included hot spot navigation and spatial audio.

✓ **Modular Building Configurator**

Created an interactive product builder with drag-and-drop 3D modules, realtime color and material changes, and backend sync. Used shader material blending for previewing textures.